A fourteen-variable consistency equation

Positive integer unknowns; exact degree 30; a 90-operation evaluator.

The only unknowns are the fourteen distinct positive integers

$$J, F, \alpha, f, g, T, X, Y, \xi, \eta, \zeta, \upsilon, \gamma, \vartheta \in \mathbb{Z}_{>0}.$$

All other symbols below are polynomial abbreviations:

$$\begin{gathered}
b = \beta J, \quad q = 1 + b, \quad C = q - F - \alpha, \quad U = Yq^2, \\
E = XY, \quad k = \eta + \zeta, \quad \ell = EYk, \quad c = \eta + Yk, \\
a = E + Y, \quad H = 3 + 4a, \quad \Delta = H + a^2, \quad L = X + ac, \\
M = q^3 b, \quad W = C(X + Kq) - Fq, \\
R = [\alpha + (q - F)b](q^2 - 1) + J(\mu + \omega q), \\
Z = M(\ell + k + \xi - kW) + Yb - Wq^3, \\
D = UL + H(U\gamma + Eq + R + 1 - k), \\
I = c^2 g, \quad Q = \Delta(f^2 - 1), \quad V = f^2(Tc - R) - c, \\
\mathcal{P} = Z^2 - M^2[1 + \ell(\ell + k)], \\
\mathcal{D} = D^2 - U^2(1 + \Delta c^2), \\
\mathcal{Q} = \upsilon^2 - Q[\upsilon^2 - \vartheta] - 1, \\
\mathcal{R} = I^2 - Q, \quad \mathcal{E} = \vartheta - V^2.
\end{gathered}$$

The single Diophantine equation is

$$\boxed{\mathcal{P}^2 + \mathcal{D}^2 + \mathcal{Q}^2 + \mathcal{R}^2 + \mathcal{E}^2 = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1, \\
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.
\end{gathered}$$

$$\begin{aligned}
\mu &= \beta - G_{951}(\rho) \\
&\quad - \rho^{960} G_{26}(\rho) - \rho^{1004}, \\
\omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda \\
&\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1004}, \\
K &= \rho^{18} + \rho^{1004} + \rho^{2020} \\
&\quad + 2\Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{gathered}
\chi_i = 1 + 2^{10}(1 + w_i + 2^8 n_i \\
+ 2^{17} p_i + 2^{25} s_i) \quad (0 \le i \le 941), \\
\pi(0) = 0, \quad \pi(i) = 1 + \sum\nolimits_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).
\end{gathered}$$

$$\begin{aligned}
\Phi(x) = &\sum\nolimits_{i=0}^{941} \chi_i x^{\pi(i)} \\
&+ 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\
&+ 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \left\lfloor \mathcal{A}/2^{636} \right\rfloor, \\
\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1 - \nu_j} \quad (0 \le j < 430), \\
\sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430), \\
m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14), \\
u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216, \\
v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).
\end{gathered}$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
```

The fixed integer $\mathcal{A}$ above is written in base 10. Lean proves that the formalized equation has a positive solution exactly when the corresponding 13-variable degree-36 equation does in every arithmetic model used by the proof. The intuitionistic PRA endpoint for this direct-square variant is in progress. The exact degree is certified by integer-polynomial calculations. The 90-operation count is for the unshifted circuit; neither count is a minimality claim.

Exact 90-operation circuit

Each numbered line defines one gate over the integers. The inputs are the fourteen unknowns and the four fixed integers above. The ASCII names `alpha, xi, eta, zeta, upsilon, gamma, vartheta, beta, mu, omega, ell, Delta` refer to their displayed counterparts; `g` is the unknown g. The last gate is zero exactly when the displayed equation holds. Intermediate gate names below are local abbreviations, not additional unknowns.

```
1.  b = J * beta
2.  q = 1 + b
3.  q2 = q * q
4.  q3 = q * q2
5.  U = Y * q2
6.  qF = q - F
7.  C = qF - alpha
8.  E = X * Y
9.  k = eta + zeta
10.  kY = Y * k
11.  c = eta + kY
12.  c2 = c * c
13.  a = E + Y
14.  g15 = 4 * a
15.  H = 3 + g15
16.  g17 = a * a
17.  Delta = H + g17
18.  g19 = a * c
19.  L = X + g19
20.  ell = E * kY
21.  M = q3 * b
22.  g25 = K * q
23.  g26 = X + g25
24.  g27 = C * g26
25.  g28 = F * q
26.  W = g27 - g28
27.  g30 = ell + k
28.  g31 = g30 + xi
29.  root_skew = k * W
30.  root_inner = g31 - root_skew
31.  root_scaled = M * root_inner
32.  g33 = Y * b
33.  root_btq3 = W * q3
34.  root_sum = root_scaled + g33
35.  Z = root_sum - root_btq3
36.  g38 = Z * Z
37.  g39 = M * M
38.  g40 = ell * g30
39.  g41 = 1 + g40
40.  g42 = g39 * g41
41.  A0 = g38 - g42
42.  g44 = qF * b
43.  rankseed = alpha + g44
44.  g46 = q2 - 1
45.  g47 = rankseed * g46
46.  g48 = omega * q
47.  g49 = mu + g48
48.  g50 = J * g49
49.  R = g47 + g50
50.  g52 = k - R
51.  r0 = g52 - 1
52.  g54 = U * gamma
53.  g55 = E * q
54.  g56 = g54 + g55
55.  shift = g56 - r0
56.  g58 = U * L
57.  g59 = H * shift
58.  D = g58 + g59
59.  g61 = D * D
60.  g62 = U * U
61.  g63 = Delta * c2
62.  g64 = 1 + g63
63.  g65 = g62 * g64
64.  A1 = g61 - g65
65.  f2 = f * f
66.  rho = f2 - 1
67.  Q = Delta * rho
68.  I = c2 * g
69.  g71 = I * I
70.  g72 = g71 - Q
71.  g74 = T * c
72.  g75 = g74 - R
73.  g76 = f2 * g75
74.  V = g76 - c
75.  y2 = upsilon * upsilon
76.  g80 = V * V
77.  g81 = y2 - vartheta
78.  g82 = Q * g81
79.  Fa = y2 - g82
80.  auxResidual = Fa - 1
81.  firstSquare = A0 * A0
82.  mainSquare = A1 * A1
83.  auxSquare = auxResidual * auxResidual
84.  strongSquare = g72 * g72
85.  firstMain = firstSquare + mainSquare
86.  firstMainAux = firstMain + auxSquare
87.  sumSquares = firstMainAux + strongSquare
88.  difference = vartheta - g80
89.  constraint = difference * difference
90.  result = sumSquares + constraint
```